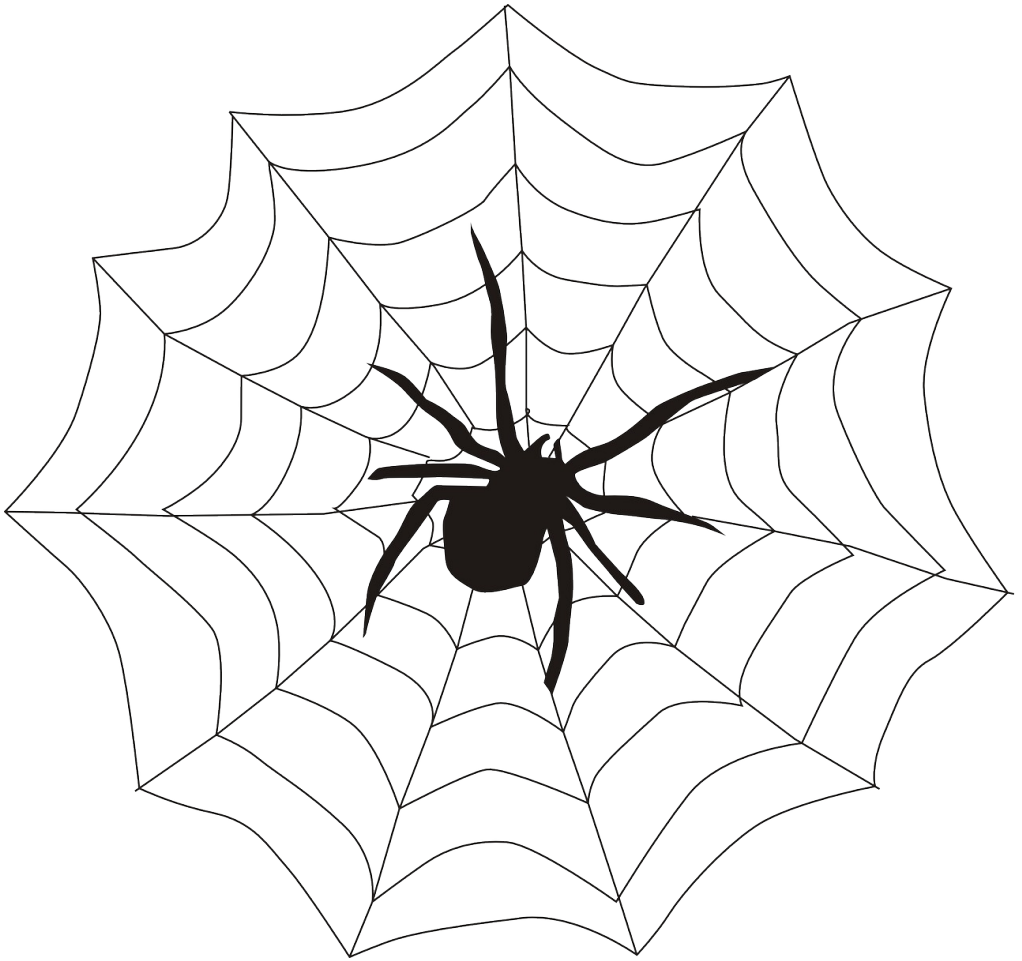
HackerFrogs Afterschool
Web App 8: Burp Suite Overview

Class:
Web App Hacking

Workshop Number:
AS-WEB-08

Document Version:
1.5

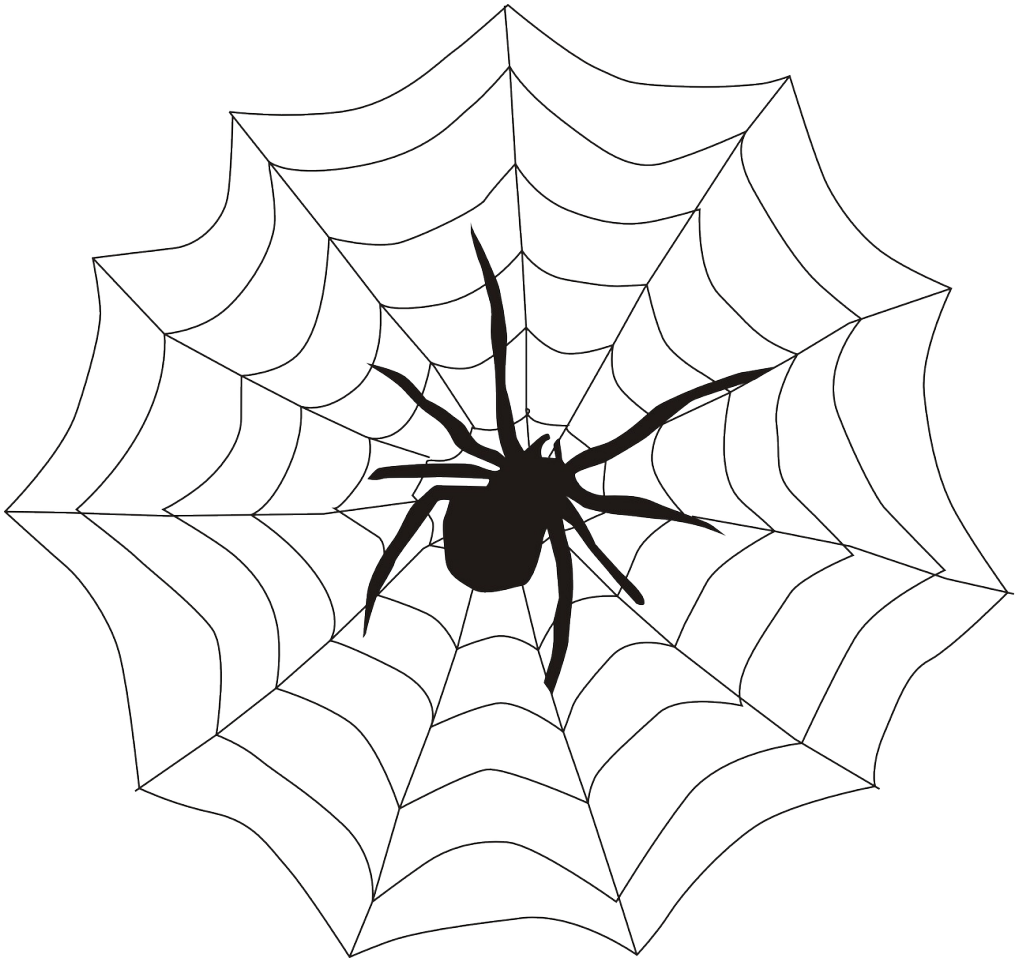
Special Requirements:
Registered account at
tryhackme.com



What We Learned In The Previous Workshop

This is the eight intro to web app hacking workshop.

In the previous workshop we learned about the following web app hacking concepts:



Blind SQL Injection: Boolean-Based

`True or False`

Boolean Based SQL Injection is a type of Blind SQL Injection where the only output that is returned after a SQL statement is whether the query returned true or false.

Blind SQL Injection: Time-Based

And we also practiced performing time-based SQL attacks, which rely on injecting a delayed response into database queries to enumerate information about the database contents

This Workshop's Topic

In this workshop, we'll be looking at Burp Suite, an industry-standard app used for web-app security testing.

Let's Learn More With TryHackMe

Navigate to the following URL:

Https://tryhackme.com/r/room/dvwa

Introduction

Burp Suite is the web app testing tool of choice for cybersecurity professionals who test web apps. It can intercept the traffic from web apps and do all sorts of stuff with the data.

What is Burp Suite?

Burp Suite captures and enables the modification of web traffic between the browser and the server, and includes the following 5 tools for web app testing:

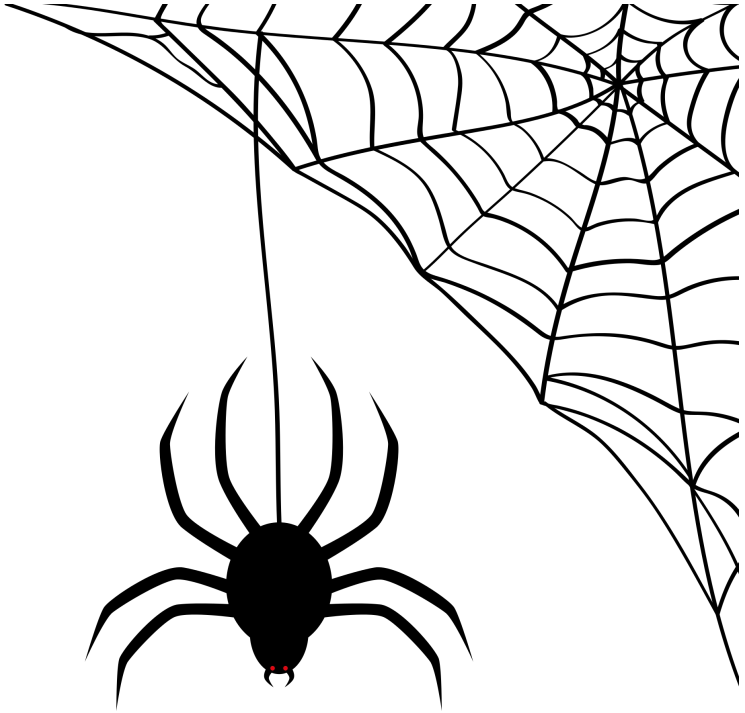
What is Burp Suite
1. Burp Proxy

The Burp Proxy server is what enables web traffic to be captured and recorded. It sits between the browser and the server, enabling traffic capture.

What is Burp Suite
2. Burp Spider

The Burp Spider is an web app indexing robot which maps out different endpoints inside a target web app.

Similar tools in the same role include Dirb, Gobuster, and Ferroxbuster



What is Burp Suite
3. Burp Intruder

Burp Intruder is a tool that can be used for different web brute forcing attacks, with modular payloads and word lists.

Similar tools include Hydra, Wfuzz, Ffuf, etc.

What is Burp Suite
4. Burp Scanner

Burp Scanner is an automated web app vulnerability scanner which tests all endpoints found in a web app for known vulnerabilities.

Similar tools include Nikto, OpenVAS, and ZAP

What is Burp Suite
5. Burp Repeater

Burp Repeater is a tool which allows for manual web request manipulation and testing.

What is Burp Suite
5. Burp Repeater

Similar tools include ZAP, Postman, and Charles Proxy.

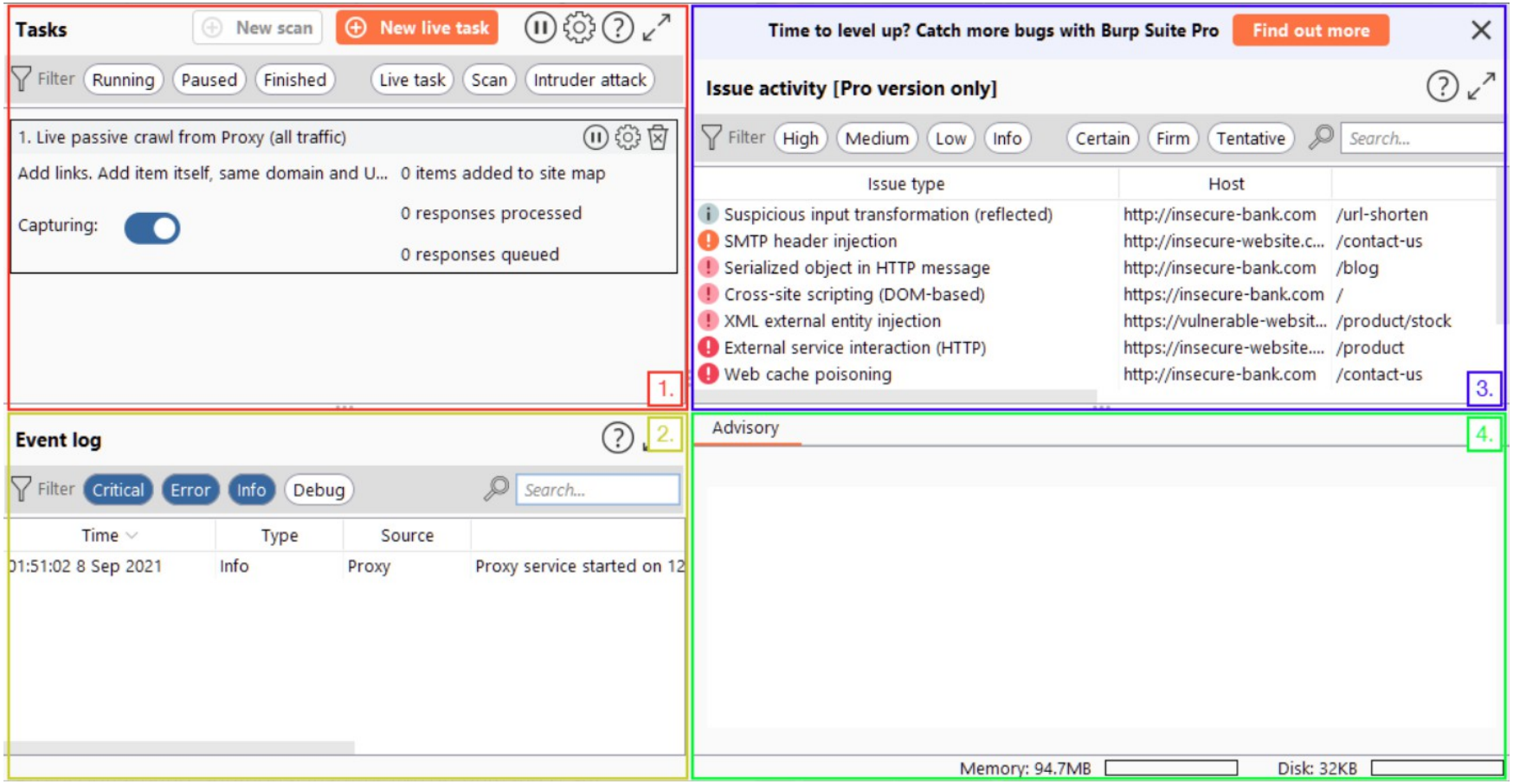
Installation

While learning Burp Suite, it's recommended to access it via virtual machines (VM), such as the TryHackMe AttackBox VM or a Kali Linux VM.

Installation

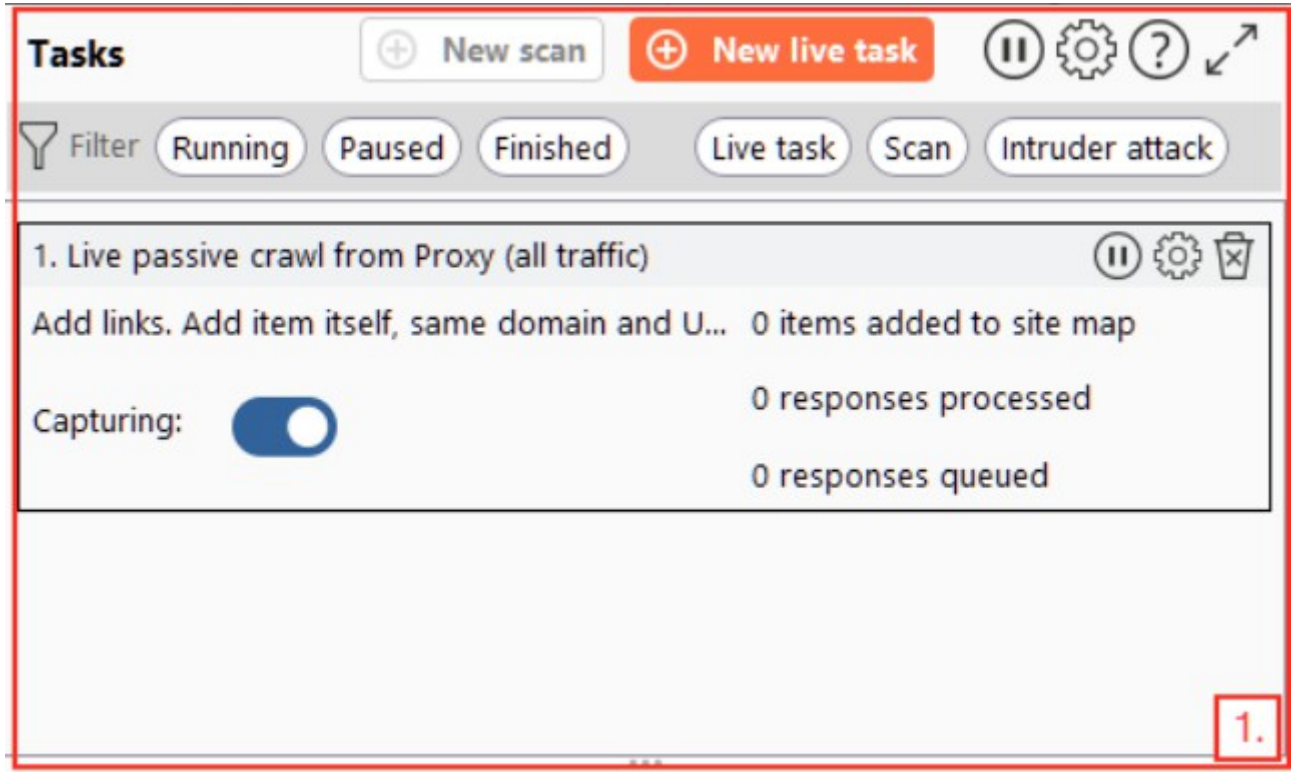
Even so, manual installation download and installation of Burp Suite is available for all major operating systems from the Portswigger company website.

The Dashboard



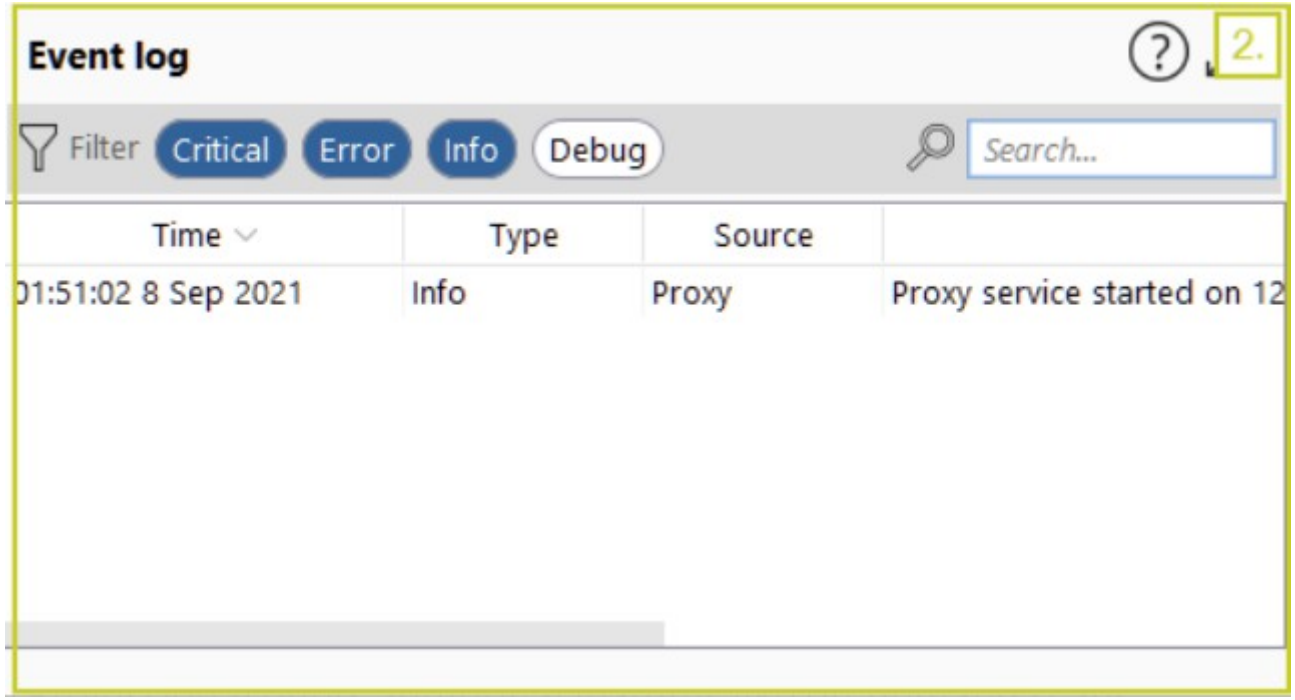
The dashboard can be subdivided into four different sections

The Dashboard



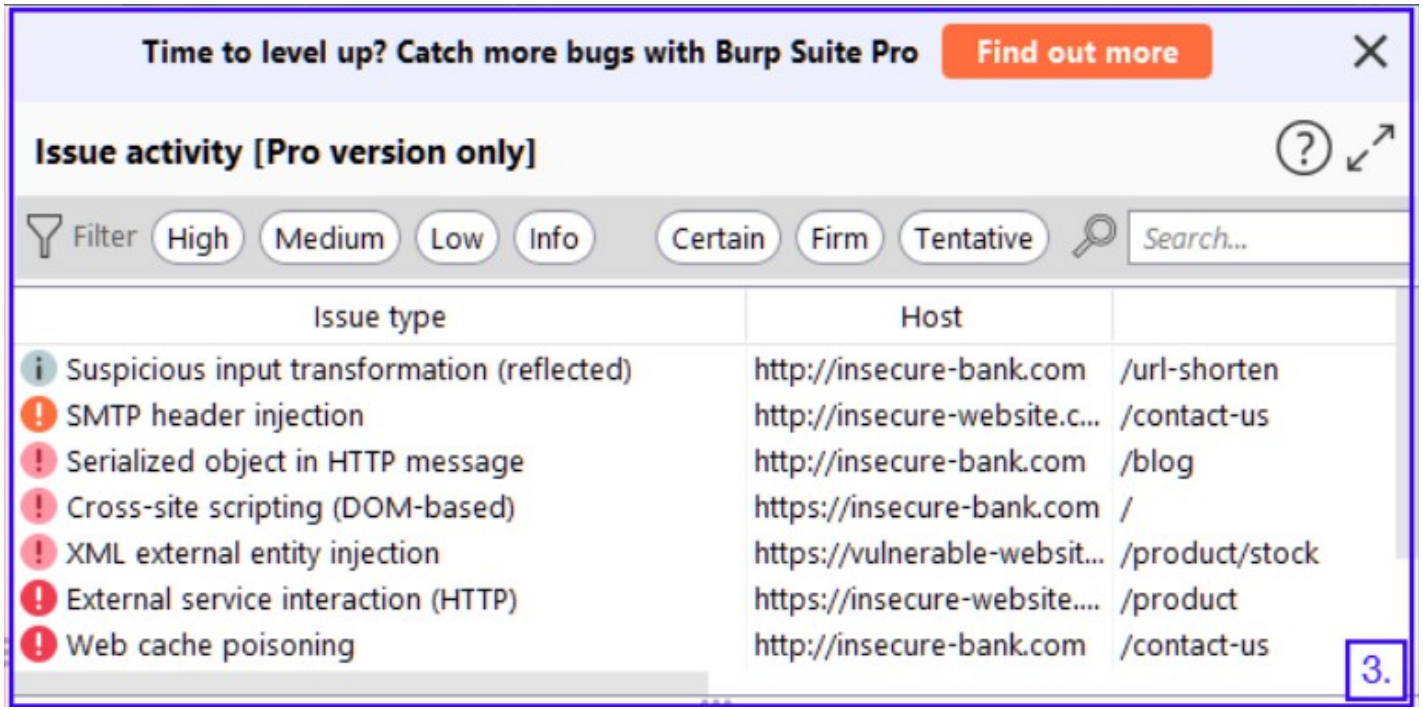
The Tasks window keeps track of visited webpages in Community edition, but in Pro edition it records scans and other tasks

The Dashboard



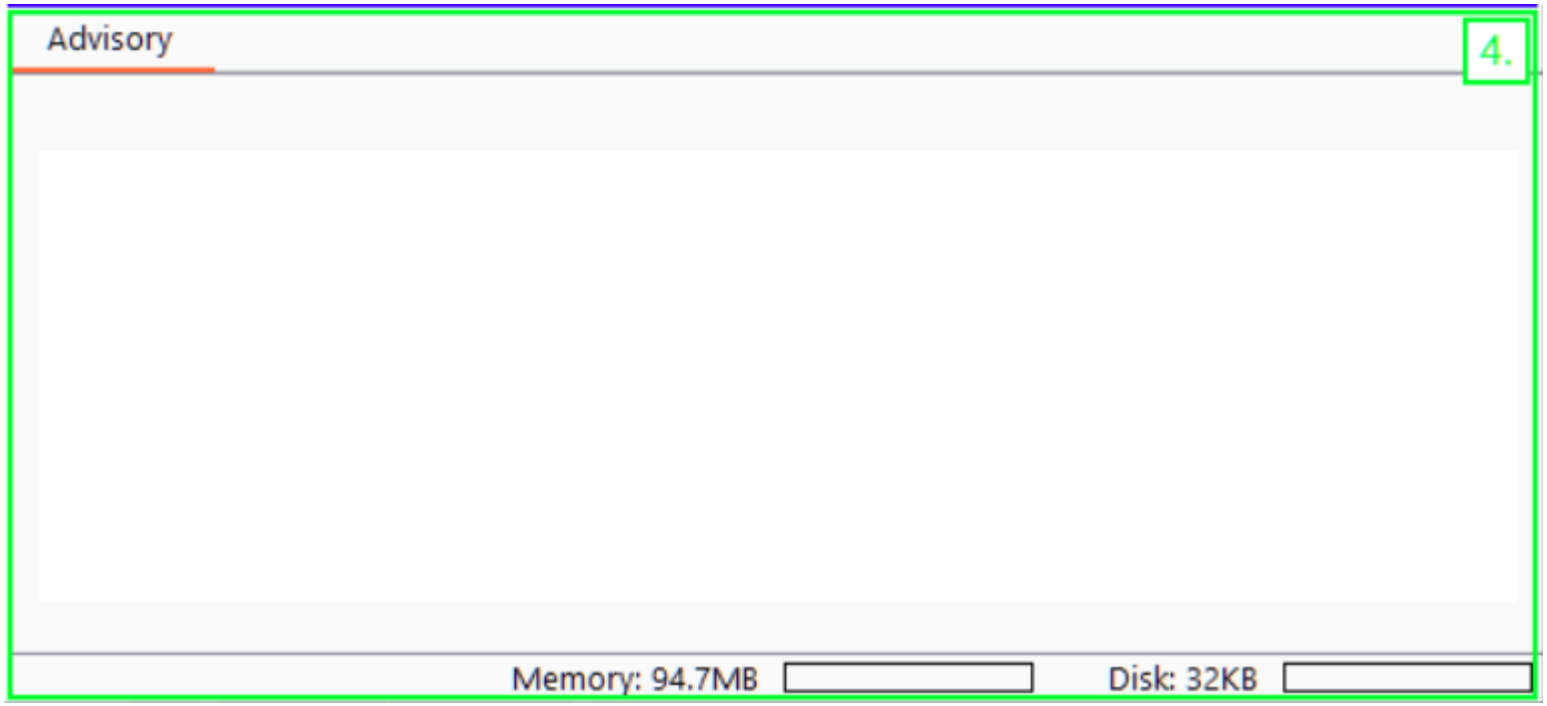
The Event Log window shows actions taken by the program for later analysis.

The Dashboard



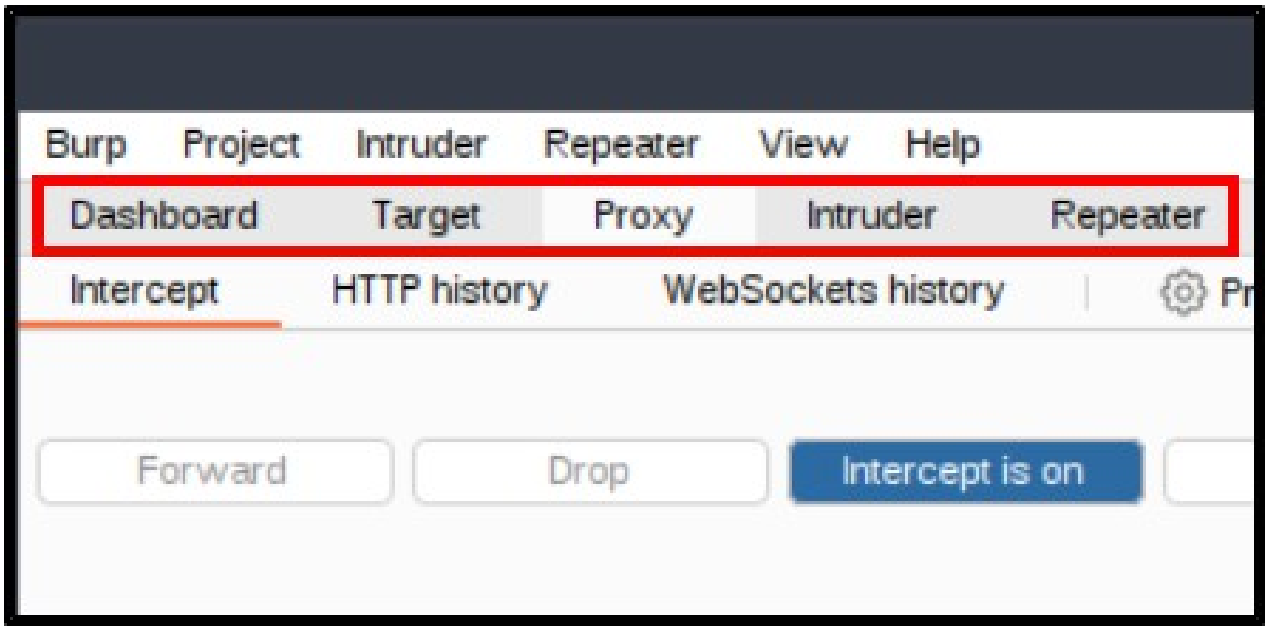
The Issue Activity window is not relevant in Community edition, but it displays vulnerabilities identified in the project scope in Pro edition.

The Dashboard



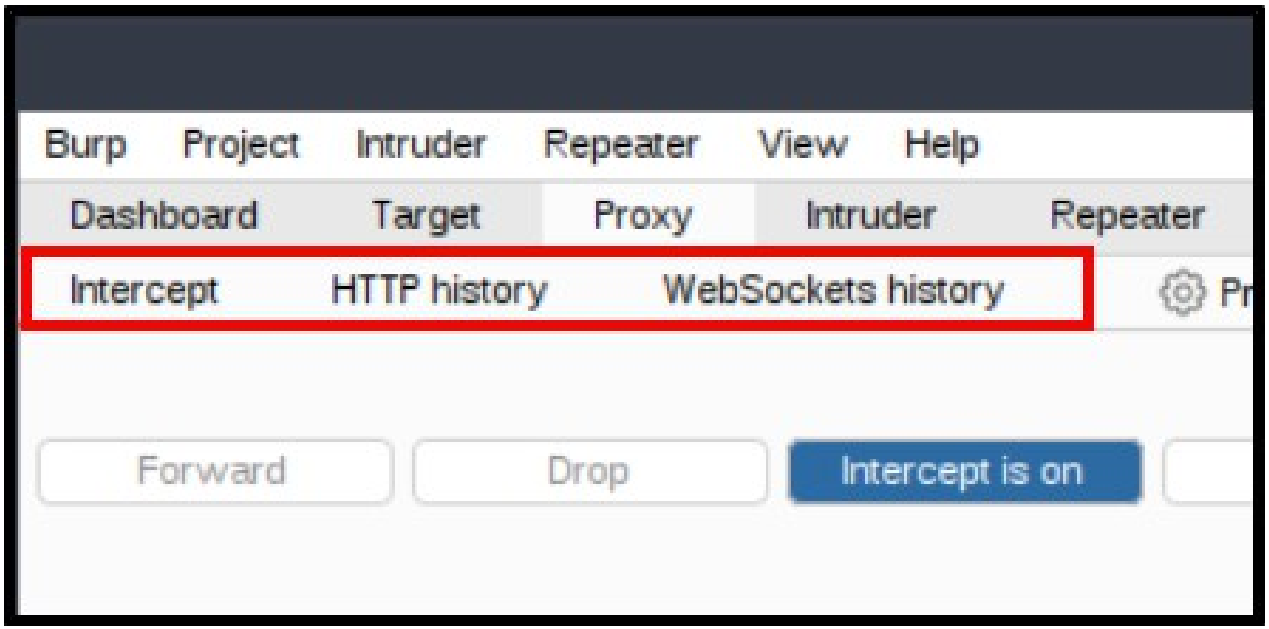
And the Advisory window provides detailed information about identified vulnerabilities, which again, is only supported in the Pro edition

Navigation



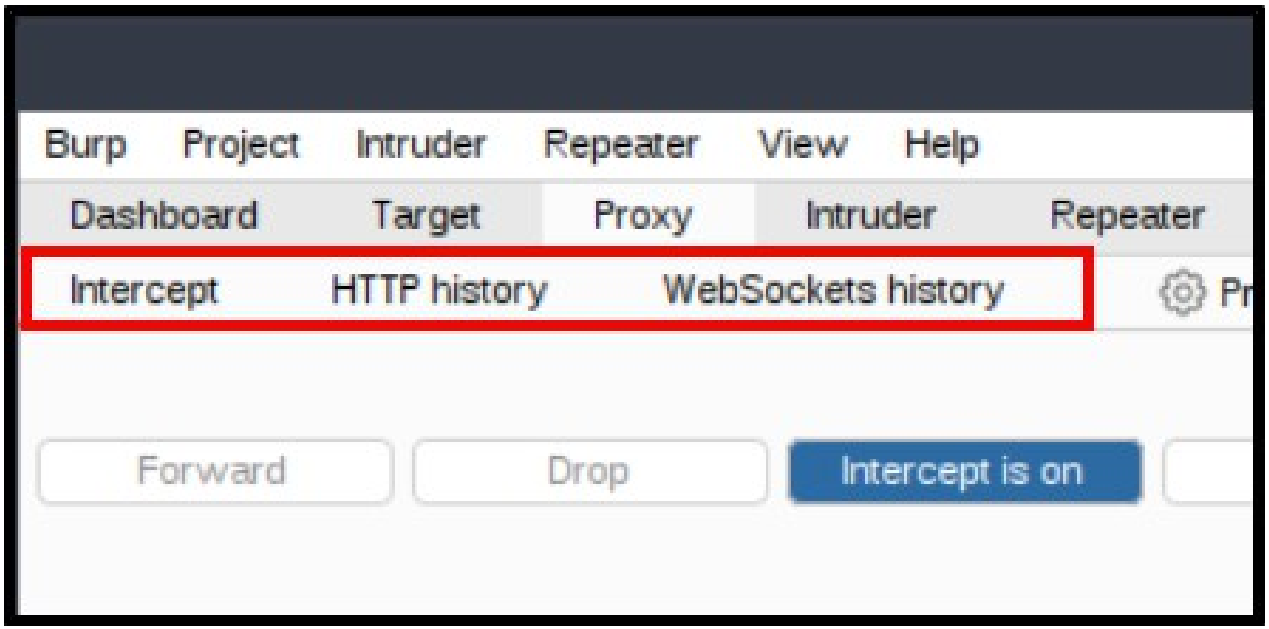
Navigating inside of Burp Suite mostly comes down to selecting the desired tab on the navigation bar, Target, Proxy, Intruder, etc.

Navigation



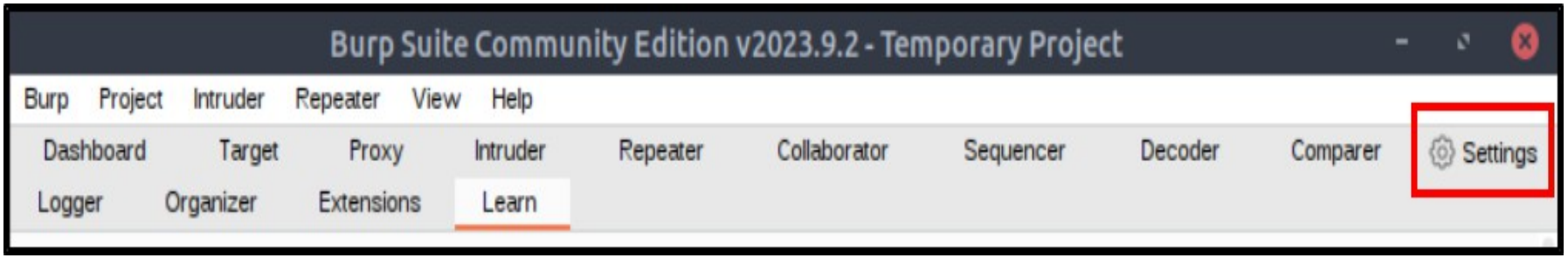
Within a specific tab, you can navigate to sub-tabs below the main tab bar.

Navigation



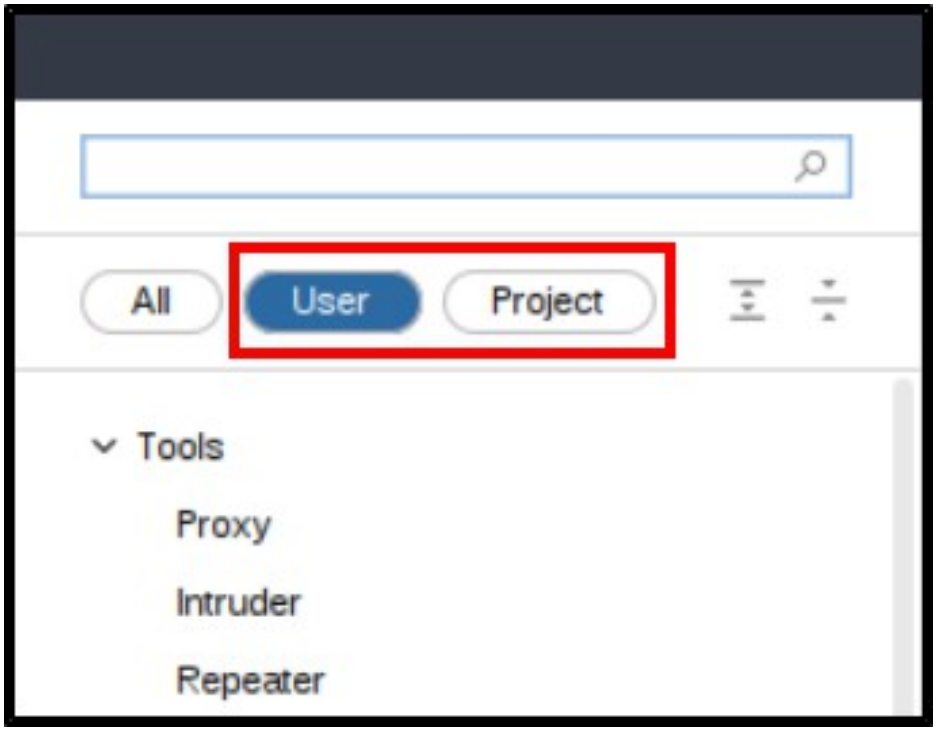
E.g., In the Proxy tab, there are the Intercept, HTTP history, and WebSockets history sub-tabs

Options



This section is actually about the Burp Suite settings page. Settings can be accessed from the settings button on the top-right corner of the UI.

Options



Once the settings page is opened, you can access User (global) settings or Project settings from their respective tabs at the top-left of the UI.

Introduction to Burp Proxy

The Burp Proxy allows web traffic to be recorded, reviewed, and replayed. One thing to take note of is the Intercept → Intercept is on button

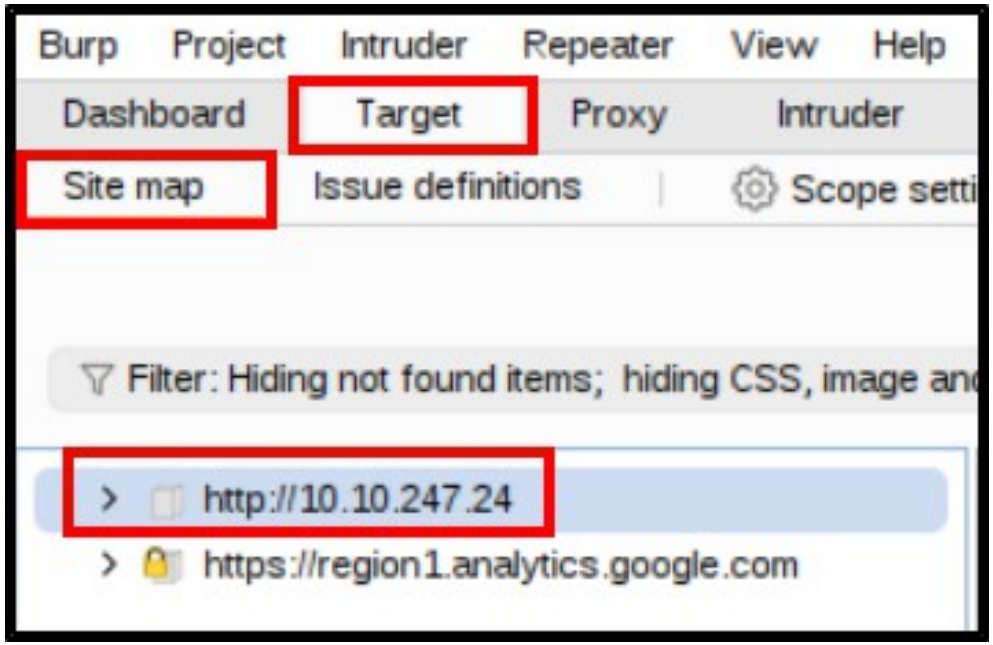
Introduction to Burp Proxy

When Intercept is on, each web request has to be manually allowed or dropped before the next can be sent.

Introduction to Burp Proxy

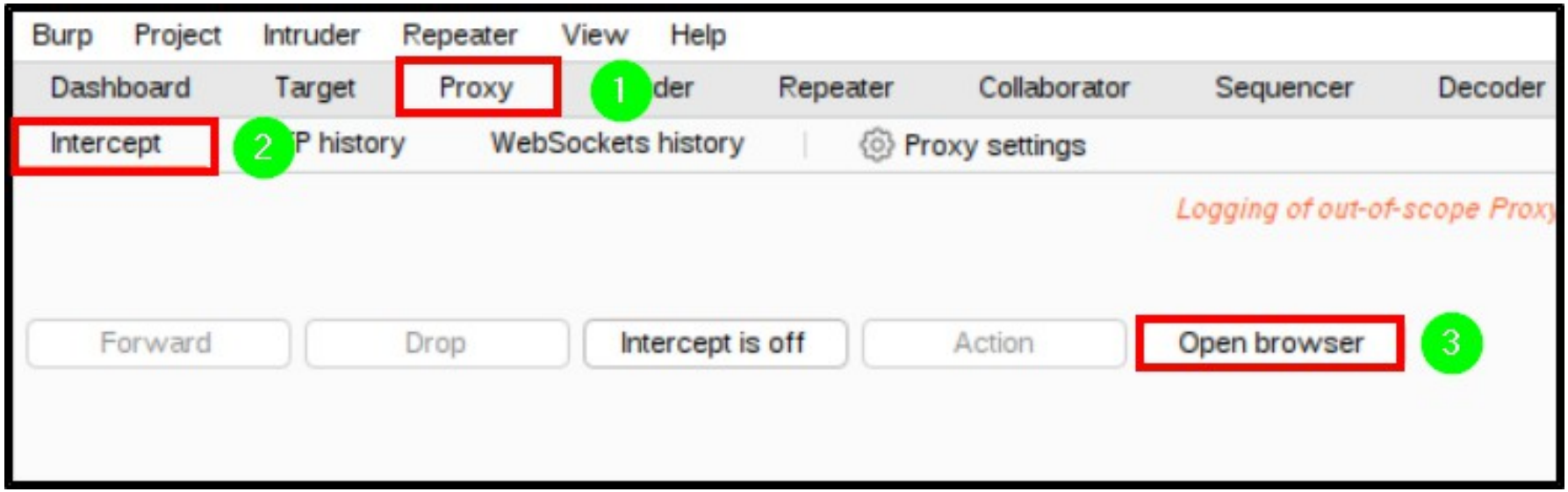
In most cases, we do not want this, and we should ensure the Intercept is on button is toggled off.

Site Map and Issue Definitions



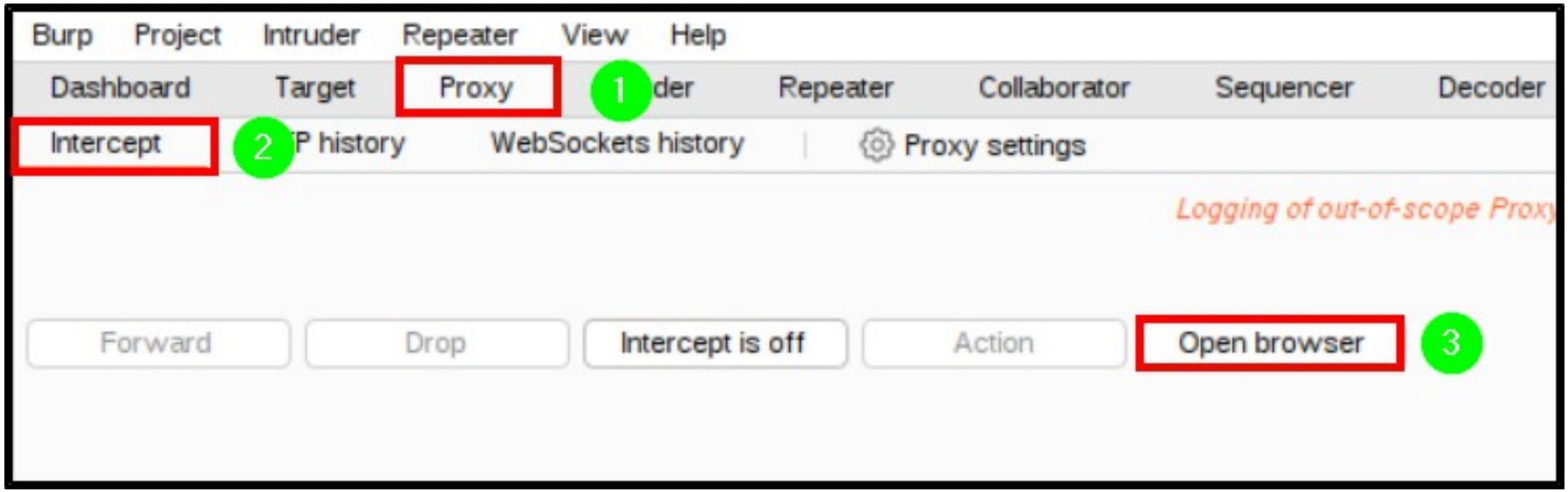
The Burp Target tab allows us to define which IP addresses and / or URLs are considered in-scope for web traffic proxy and capture.

The Burp Suite Browser



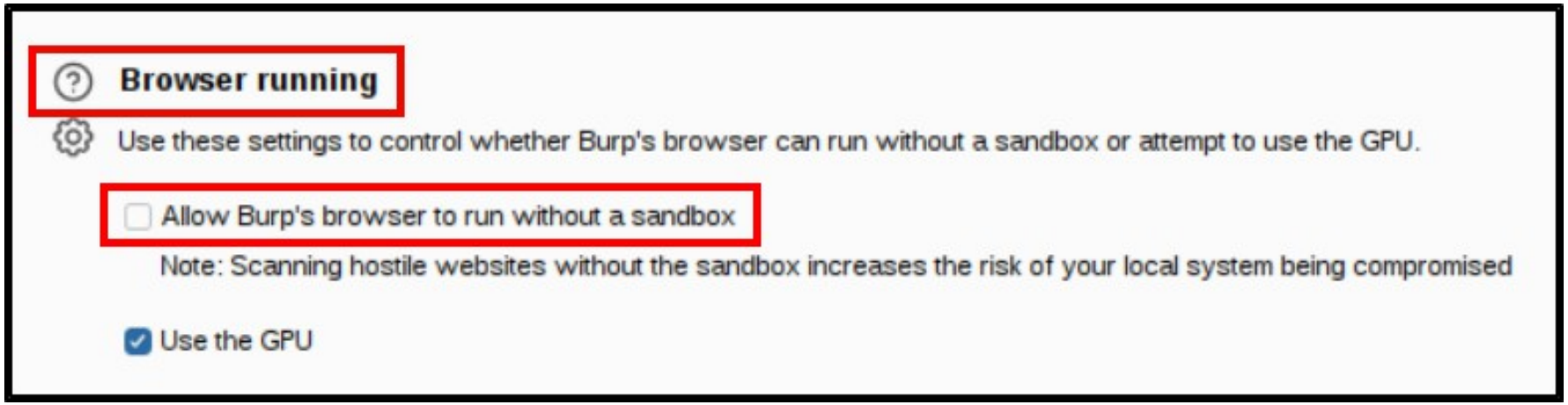
Using the Burp Suite Browser to proxy web traffic is the more convenient option over using regular web browser proxies.

The Burp Suite Browser



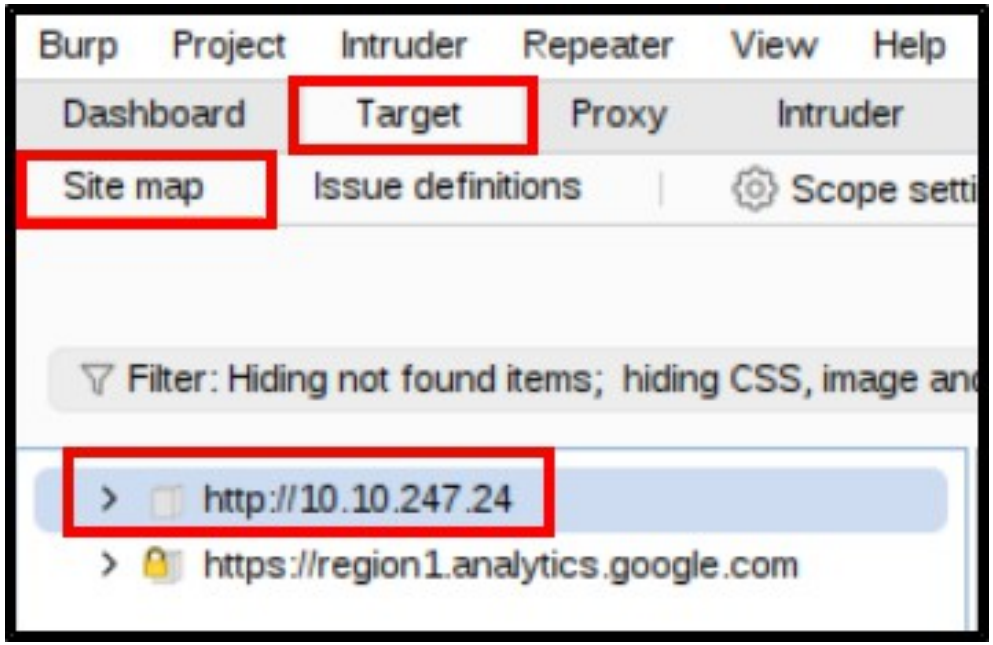
It can be accessed under the Proxy tab, then the Intercept sub-tab, then the Open browser button.

The Burp Suite Browser



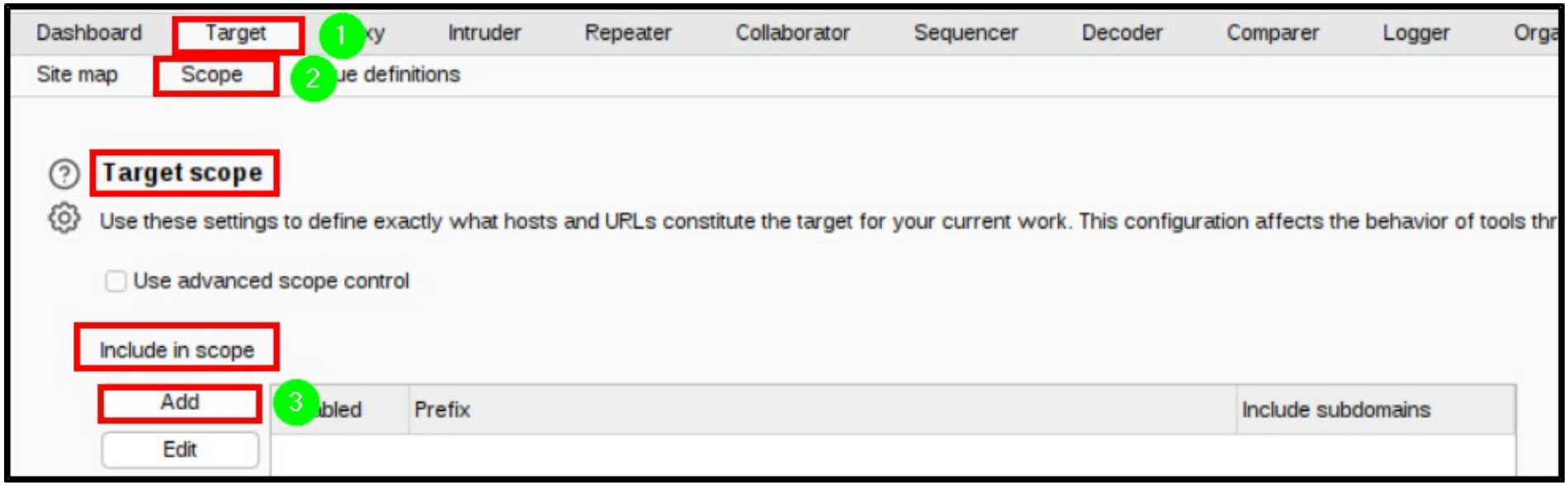
In the AttackBox, you'll have to go into Settings → Burp's browser → Browser running → Allow Burp's browser to run without a sandbox.

Scoping and Targeting



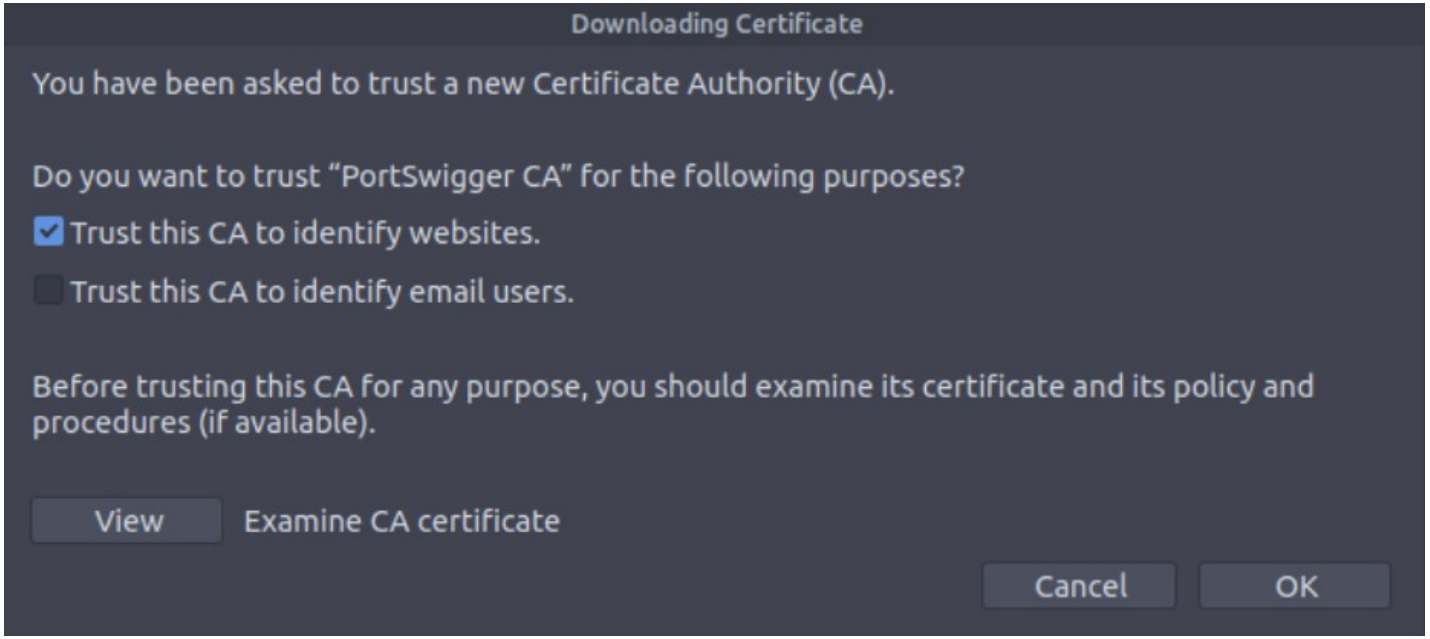
In most cases, we want to set a scope for our testing, and exclude all traffic that is outside the URL / IP address of the scope.

Scoping and Targeting



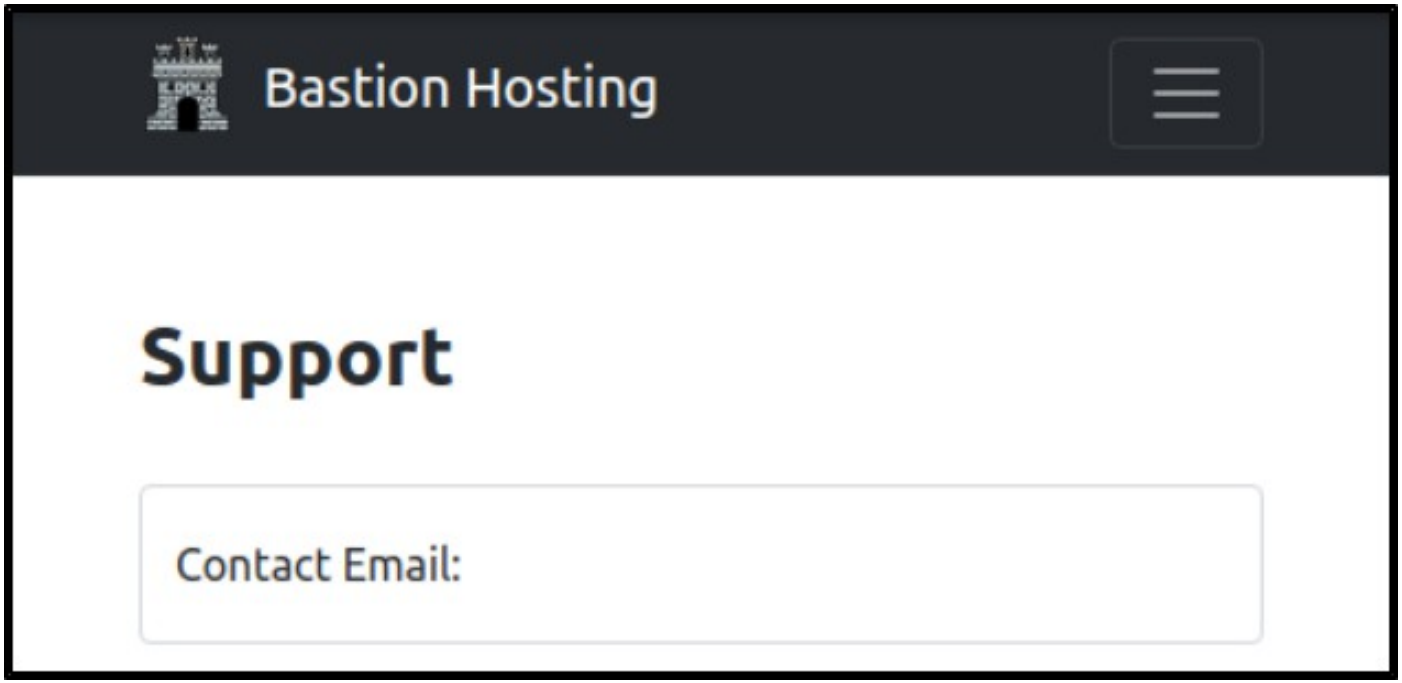
We can manually set a scope by using the Target → Scope → Target scope → Include in scope → Add button.

Proxying HTTPS



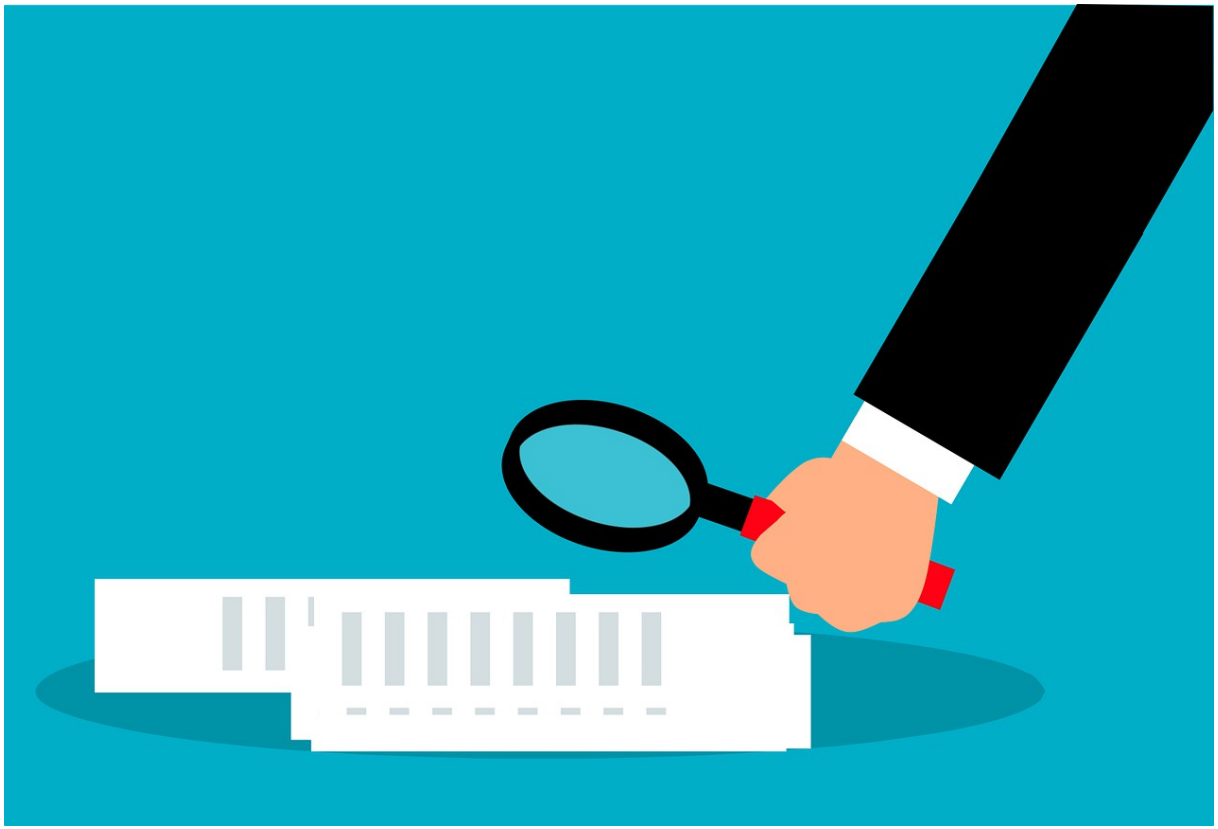
One reason why it's preferable to use Burp's Browser to proxy web traffic is because it avoids some setup steps such as downloading and configuring HTTPS certificates.

Example Attack



Our last task in this room is to perform a web attack on the room's associated web app.

Summary



Let's review the web exploitation concepts we learned in this workshop:

Burp Suite

Burp Suite is the industry-standard software framework for testing web apps. It includes many different tools, but the Burp tool that enables the rest of the framework is...

Burp Proxy

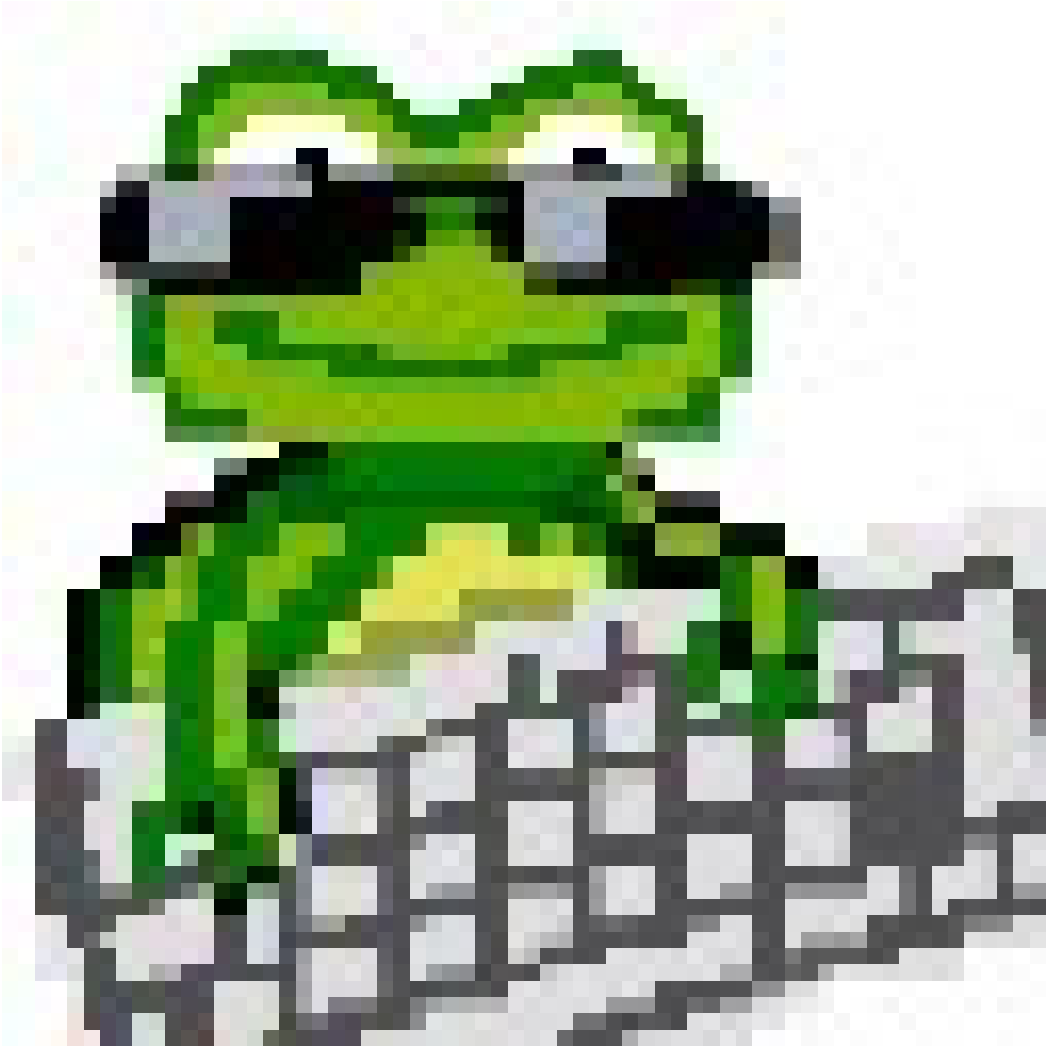
The Burp Proxy server is what enables web traffic to be captured and recorded. It sits between the browser and the server, enabling traffic capture.

What's Next?

In the next HackerFrogs Afterschool web app hacking workshop, we'll learn continue learning about Burp Suite, more specifically, the Burp Repeater tool.

Extra Credit

Looking for more study material on this workshop's topics?

See this video's description for links to supplemental documents and exercises!

Until Next Time, HackerFrogs!